

Vinay Srigadi

Bowenpally, Hyderabad

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Professional Summary

AI/ML Engineer and Full-Stack Developer specializing in generative deep learning (VQ-VAE, Transformers, LSTM) and production-grade systems (React, FastAPI, Flutter, Docker). Deployed a 30M-parameter music-to-dance AI achieving 85% beat accuracy on consumer hardware; presented original LSTM choreography research at NCNTAIA-2025 (national conference). CGPA 9.49 — seeking roles at the intersection of applied AI and scalable product engineering.

Internship Experience

Tech Lead *Viswam.ai*

IIT Hyd, Gachibowli
July 2025 – Sept 2025

- Led full-stack development of the Corpus app (React), delivering multi-lingual support across 3 languages; redesigned task navigation flow reducing annotator clicks per task from 9 to 4, improving throughput for 20+ active users.
- Designed JWT-secured RESTful API with role-based access control, decoupling auth logic from UI layer and enabling safe multi-tenant access across the platform.
- Migrated intern tracking dashboard from Streamlit to React, cutting average page load time from 4s to under 800ms and supporting 50+ concurrent users without backend changes.
- Technologies: React, Flutter, Dart, Streamlit, REST APIs, Client-Server Architecture, UI/UX Design, Authentication Security.

AI/ML Intern

Presented at NCNTAIA-2025 Conference

IEEE Hyderabad
May 2025 – June 2025

- Designed a sequence-to-sequence LSTM pipeline to synthesize beat-conditioned 3D skeletal motion from raw audio without using any pre-trained motion priors or pose databases.
- Designed a multi-modal data pipeline using MediaPipe for skeleton extraction and Librosa for audio feature engineering, processing 100+ video clips to generate synchronized training pairs.
- Achieved 38.56% PCK on a custom LSTM-based motion generation model — a strong baseline for novel choreography synthesis without pre-trained motion priors.
- Technologies: Python, TensorFlow, Keras, MediaPipe.

Projects

AI-Based Music-to-Dance Generation System (Music2Dance AI)

2026

- Architected an end-to-end generative AI platform trained on 1,510 professional AIST++ dance sequences, mapping 140-dim audio features to beat-synchronized 2D choreography via VQ-VAE + Transformer.
- Trained VQ-VAE (512-code discrete codebook, 128-dim latent) achieving reconstruction loss of 0.0013 with 100% codebook utilization — zero dead codes or mode collapse.
- Developed 6-layer causal Transformer decoder (30M parameters) with audio-conditioned autoregressive generation; best validation loss: 2.5525.
- Engineered 5-layer anti-freeze diversity system achieving 50–70% token diversity across 50+ test generations; 85% beat detection accuracy via Librosa BPM analysis.
- Deployed production-grade three-tier platform: Next.js 14 (Vercel) + FastAPI/Docker (Render) + MongoDB Atlas; JWT-secured with async processing, generating 720×720 30FPS MP4 videos in 15–30 seconds on consumer hardware.
- Designed 6 formal algorithms and 9 UML diagrams; cited 30 academic references including Vaswani et al. (2017), Li et al. AIST++ (2021), and Siyao et al. Bailando (2022).
- Technologies: Python, PyTorch, FastAPI, Next.js, MongoDB Atlas, FFmpeg

Plant Identification Mobile App

2024

- Deployed a TFLite CNN model for offline plant identification in Flutter, achieving sub-200ms inference latency with 85%+ classification accuracy across 30 plant species.
- Technologies: Flutter, Dart, TensorFlow Lite

Mental Health Mobile App

2023

- Built and deployed a full-stack mental health app (Flutter + Flask + MongoDB Atlas) with Mistral-7B-powered conversational AI, mood tracking, and appointment scheduling; served as capstone showcasing end-to-end LLM deployment on commodity hardware.
- Technologies: Flutter, Flask, MongoDB Atlas, Mistral-7B

Technologies

AI/ML: PyTorch 2.1, TensorFlow/TFLite, Keras, Hugging Face, MediaPipe, OpenCV, Librosa, FFmpeg

Full-Stack: React, Next.js 14, Flutter/Dart, FastAPI, Flask, TypeScript, TailwindCSS

Data & Infra: MongoDB Atlas, MySQL, Docker, AWS, Vercel, Render, CI/CD, Git

Languages: Python, Dart, JavaScript/TypeScript, Java, C

Education

Master of Computer Applications, Koneru Lakshmaiah Education Foundation

Expected 2026

- CGPA: 9.66

B.Sc.(Computer Science), Kishinchand Chellaram College, HSNC University

2021 – 2024

- CGPA: 9.61

Certificate & Achievements

- **AWS Certifications:** Cloud Practitioner, Developer – Associate, Machine Learning Engineer – Associate.
- **Gold Medal – 1st Prize**, 13th All India Children & Youth Art Competition, **Andhra Pradesh State Creativity & Culture Commission**, Vijayawada (Feb 2026)
- Certificate of Presentation, **NCNTAIA-2025**, for “*Choreography AI: Generating New Dance Steps*” — awarded by Anusandhan National Research Foundation
- Completed **AI with Machine Learning** workshop — Skilligence Edtech, BITS-Pilani Hyderabad Campus
- **Participation Certificate:** Qualified for Final Round of RAMpage V2.5 24-Hour Hackathon — KLH University, GeeksforGeeks, IEEE, and Algorand Blockchain Club
- Certificate of Participation – **SCIENT SPARK Startup Launchpad** (Jan–Mar 2025)
- **Virtual Internship in AI/ML** – V VUBE Software Solutions (Feb 2025)
- **Best Performance Award** – MCA Program
- **School Rank 1** – Secondary School Certificate (Grade 10 board), 83.8%